

# TIC 224215371

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

|                                |                                 |                               |          |
|--------------------------------|---------------------------------|-------------------------------|----------|
| BLS EPHEMERIS                  |                                 | CATALOG CROSS-MATCH (novelty) |          |
| Period                         | 5.493091 d (ALIASED, see below) | VSX                           | no match |
| -> corrected period:           | 10.986182 d                     | ASAS-SN                       | no match |
| Epoch (t0)                     | 2449.2516 BTJD                  | Gaia eclipsing-binary table   | no match |
| Duration                       | 0.0840 d                        | Gaia variable classification  | no match |
| Depth                          | 1.431 % (14,311 ppm)            | TESS EB catalog (Prsa+2022)   | no match |
| SNR                            | 52.2                            | ExoFOP TOI list               | no match |
| INDEPENDENT CHECKS (f3i gates) |                                 | Verdict: NOVEL                |          |
| Eclipses observed              | 5 (gate >= 3)                   |                               |          |
| Robust depth ratio             | 0.88 (gate >= 0.50)             |                               |          |
| Robust depth sigma             | 34.5 (gate >= 7.0)              |                               |          |
| Odd/even z-score               | 6.36 (ALIAS FLAGGED)            |                               |          |
| Secondary eclipse              | detected, phase 0.49, 6.7 sigma |                               |          |

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

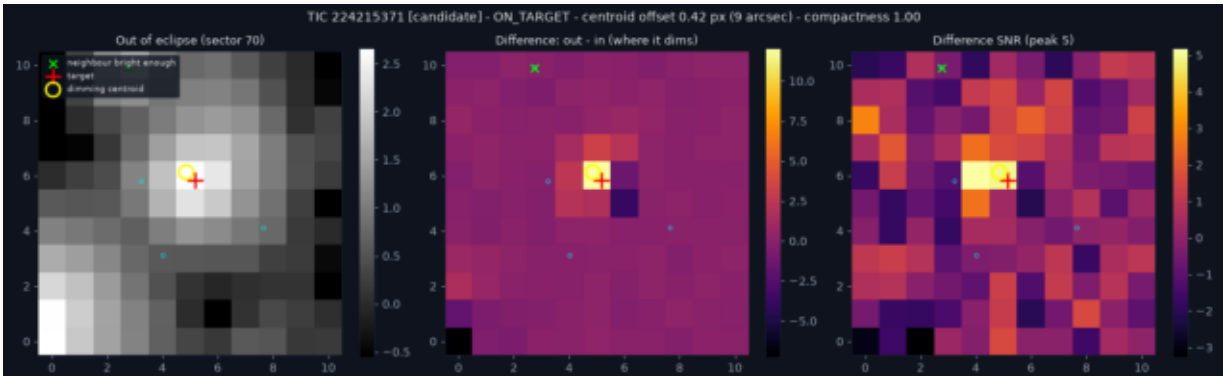
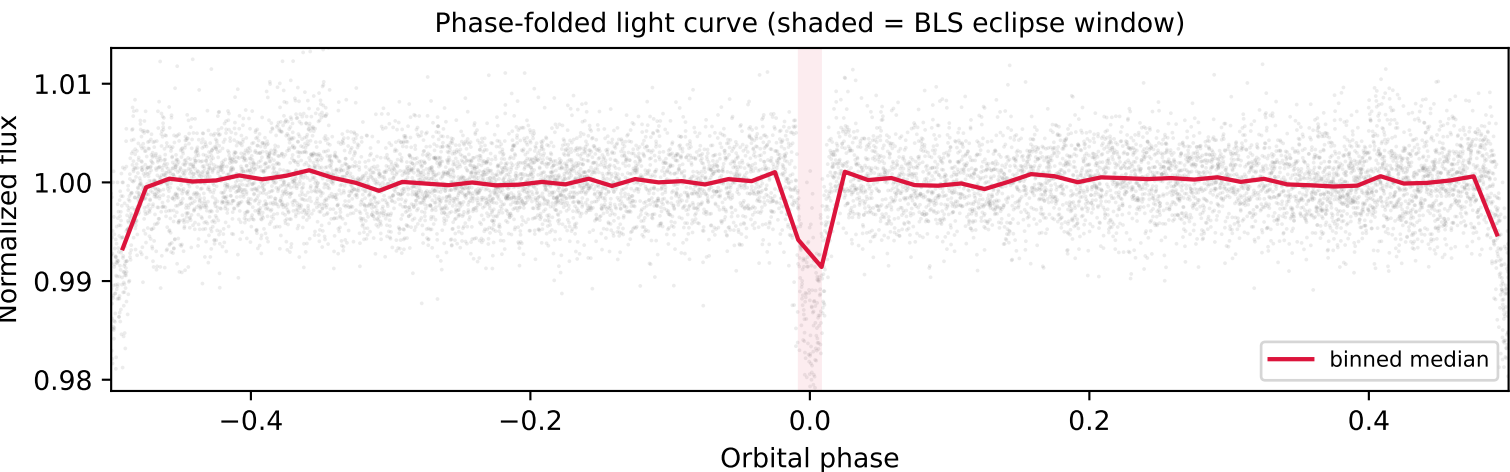
Sector 70 light-curve dip in this sector: 28.8 sigma t0 refined by 0.0025 phase

Difference-image peak SNR 5.18 (gate >= 5.0) Compactness 1.00 (gate >= 0.50)

Dimming centroid offset 0.42 px (9 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 224217503

4.32 px away, dTmag +2.56 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 6.36 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 10.986182 d, in any write-up.